Constrained Extrema and Lagrange Multipliers

Optimal Control, Homework 3

Arshia Goshtasbi • MSc Electrical Engineering (Control), Iran University of Science and Technology • Spring 1404 (2025)

English summary of my original Persian report, which is kept as a PDF in this folder.

The original is handwritten (tablet) in Persian. The problems come from Chapter 4 of Kirk, Optimal Control Theory.

Problem 4-22 (static constrained extremum). I extremised $f = y_1^2 + y_2^2 + y_3^2$ subject to $y_1 + y_2 + y_3 = 5$ and $y_1^2 + y_2^2 + y_3 = 9$. I formed the augmented function with two multipliers and set its differential to zero, then examined the case where $y_1^\star = y_2^\star$ and $p_2^\star = -1$. This gives the candidates $(2, 2, 1)$ with $f = 9$ and $(-1, -1, 7)$ with $f = 51$. The other case ($p_1^\star = 0$) has a negative discriminant and no real solution.

Problem 4-24 (differential constraints). For $J = \int(\lambda + w_3^2)\,dt$ with $\dot{w}_1 = w_2$ and $\dot{w}_2 = w_3$, I wrote the augmented integrand $g_a = \lambda + w_3^2 + p_1(\dot{w}_1 - w_2) + p_2(\dot{w}_2 - w_3)$. The three Euler–Lagrange equations give $p_1^\star$ constant, $\dot{p}_2^\star = -p_1^\star$ and $p_2^\star = 2w_3^\star$.

Problem 4-25 (isoperimetric constraint). To minimise $\int_0^1 (\dot{x}^2 + t^2)\,dt$ with $x(0) = x(1) = 0$ and $\int_0^1 x^2\,dt = 2$, I introduced z with $\dot{z} = x^2$. The Euler–Lagrange equations give $\ddot{x} = Cx$ with constant C. Only $C = -n^2\pi^2$ admits a non-trivial solution satisfying both end conditions, and the integral constraint fixes the amplitude: $x^\star(t) = 2\sin(n\pi t)$.